



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-11920

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Mon, May 20, 2024 at 11:26 PM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

21-May-2024

Dear Mr. Khare:

The manuscript entitled "Deep Reinforcement Learning-based Optimization Method for D2D Communication Energy Efficiency in Heterogeneous Cellular Networks" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Deep Reinforcement Learning-based Optimization Method for D2D Communication Energy Efficiency in Heterogeneous Cellular Networks

AUTHORS: Pan, Ziyu; Yang, Jie

ABSTRACT: Facing the challenge of exponentially increasing data traffic on communication networks brought by the 5G era, this paper focuses on how to apply deep reinforcement learning (DRL) techniques to solve the problem of optimizing the energy efficiency of D2D communications in heterogeneous cellular network environments. We innovatively propose a joint resource allocation scheme based on multi-intelligent deep reinforcement learning, which enables D2D devices to intelligently switch between license-free and optimal license bands, and adjust the transmit power in real-time to maximize the energy efficiency improvement. In this paper, a multi-intelligent deep reinforcement learning framework is designed to enable D2D users in heterogeneous networks to make collaborative decisions and dynamically adjust their communication strategies according to real-time network status and environmental changes. In this paper, a deep Q-network model with Graph Attention Network (GAT) as the core structure is constructed, which is able to cope with the complexity and diversity of network states, and learn and execute optimal resource allocation strategies. In this paper, we propose a targeted loss function design that balances the optimization goal of D2D communication energy efficiency with network stability and long-term gains. Through rigorous simulation experiments, this paper verifies that the DRL-based approach can significantly improve the energy efficiency of D2D communications in heterogeneous cellular networks in real-world scenarios, while ensuring the stability of the quality of service (e.g., rate, delay, and resource utilization).

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